PETER MUSILA

Nairobi County, Kenya | +254 719443774

| musiladev@gmail.com

LinkedIn: linkedin.com/in/peter-musila | GitHub: github.com/peter-kiilu

---

Professional Summary

Innovative Software Engineer and Data Science specialist focused on building intelligent systems and scalable digital products. Proven ability to architect full-stack applications, implement AI-powered solutions (RAG, LLMs, agentic workflows), and transform raw data into actionable insights. Experienced in leading technical initiatives, collaborating with cross-functional teams, and delivering measurable impact. Passionate about leveraging technology to solve complex, real-world problems.

Technical Skills

- **Full-Stack Development:** React (TypeScript), Python (Flask), Node.js, PostgreSQL, RESTful API Design, System Architecture
- **AI & Data Science:** Retrieval-Augmented Generation (RAG), LangChain, Vector Databases (Chroma), Machine Learning, Exploratory Data Analysis (EDA), Data Preprocessing & Cleaning, Prompt Engineering, AI Agents & Automations
- **Cloud & DevOps:** Docker, Kubernetes, CI/CD Pipelines, Cloud-Native Development

Certifications

- **Building AI Agents and Automations** – Build Fast With AI (February 2026)
- **AI Literacy** – IBM Student Ambassador (March 2026)
- **Analyzing and Visualizing Data with PromptBI** – PromptBI (Nov 2025)
- **Exploratory Data Analysis** – Martial Jenga Labs (Jan 2025)
- **Data Preprocessing, Cleaning, Wrangling & Presentation** – Martial Jenga Labs (Jan 2025)

Professional Experience

AI Software Developer | Tech Innovators Network (THiNK) January 2026 – March 2026 | Embu, Kenya

- Built and deployed a **RAG chatbot (BETA)** using Python, LangChain, ChromaDB, and PostgreSQL**,** indexing 290+ website documents and linked pages to improve knowledge discovery.
- Automated content ingestion and semantic retrieval, achieving 95% accuracy**,** reducing response time to 2 seconds**,** and saving 45 hours of manual work weekly**.**
- Developed a responsive React frontend with secure authentication and session management**,** enabling context-aware user interactions for 20 active users**.**

Founder and Leader | Responsible Computing Young Innovators Club, University of Embu

January 2024 – Present | Embu, Kenya

- Spearheaded community-driven initiatives in **ethical AI** and responsible computing, mentoring student developers.
- Coordinated technical training sessions and innovation challenges, resulting in a **60% increase in student engagement**.

Product Developer (Internship) | MarketForce

July 2024 – February 2025 | Nairobi, Kenya

- Contributed to the design and development of scalable digital products aligned with business requirements.
- Participated in system design discussions and the implementation of **distributed architectures**; supported **cloud-native development** workflows using Docker, Kubernetes, and CI/CD practices.
- Assisted in testing, debugging, and performance optimization for production-ready systems.

ICT Support Officer (Internship) | University of Embu

May 2025 – August 2025 | Embu, Kenya

- Provided technical support for system maintenance, hardware/software configuration, and network troubleshooting to ensure operational continuity.

Education Bachelor of Science in Information Technology (Expected Graduation: Sept 2026)University of Embu, Kenya

- **Relevant Coursework:** Software Engineering, Artificial Intelligence, Data Structures & Algorithms, Database Systems, Cloud Computing.
- **Honors:** Innovation Leadership Recognition (2024)

Leadership & Community Involvement

- **Member & Event Organizer** – Google Developer Groups (GDG) on Campus, University of Embu (2022 – Present)
  Organized major tech events including the THiNK Botathons (AI/ML Hackathon), ICP Hubs Kenya (Blockchain), Moringa School Cybersecurity Bootcamp and Agentic AI Workshop, Droidcon X Fluttercon Ke26.**Innovation Awardee** – Recognized for leadership and project development at national and regional innovation forums.

Languages

- **English:** Full professional proficiency| **Swahili:** Fluent